

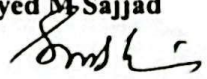


Air University
Mid Semester Examinations: Spring 2026

Student ID: _____

Subjective Part
(To be solved on Answer Books only)

Subject: Penetration Testing
Class: BS CyS Morning F-23
Section(s): Morning F-23 (A & B)
Course Code: CY-433

Time Allowed: 120 Minutes
Max Marks: 50
FM's Name: Dr Syed M Sajjad
FM's Signature:- 

INSTRUCTIONS

- Attempt responses on the answer book only.
- Nothing is to be written on the question paper.
- Rough work or writing on question paper will be considered as use of unfair means.
- Tables / calculators are not allowed.

Given

Question 1	CLO :01	GA :01	Domain level : Understand (C2)	Marks = 7*5=35
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- A. A financial organization wants to test the security of its online banking platform and internal network infrastructure before launching a new digital service.
1. Identify and explain two suitable types of penetration testing that should be conducted in this situation.
 2. Compare Black Box, White Box, and Grey Box penetration testing approaches and recommend the most suitable approach for this organization.
- B. A security consulting firm follows a structured methodology during penetration testing engagements.
- a) Describe the typical phases of a penetration testing methodology.
 - b) Explain the importance of reconnaissance and enumeration in the early stages.
 - c) Discuss how the results of these phases influence later exploitation activities.

Question 2	CLO :04	GA :04	Domain level : Creat3 (C6)	Marks = 7+8=15
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A penetration testing team is hired to assess the security of a university network infrastructure. Before starting the engagement, both the university and the penetration testing team must agree on specific conditions and guidelines.

1. Develop a comprehensive definition of Rules of Engagement (RoE) in the context of penetration testing and explain its purpose in a security assessment.
2. Design a Rules of Engagement framework for this penetration testing engagement by proposing four essential elements that should be included in the RoE document and explaining their significance.